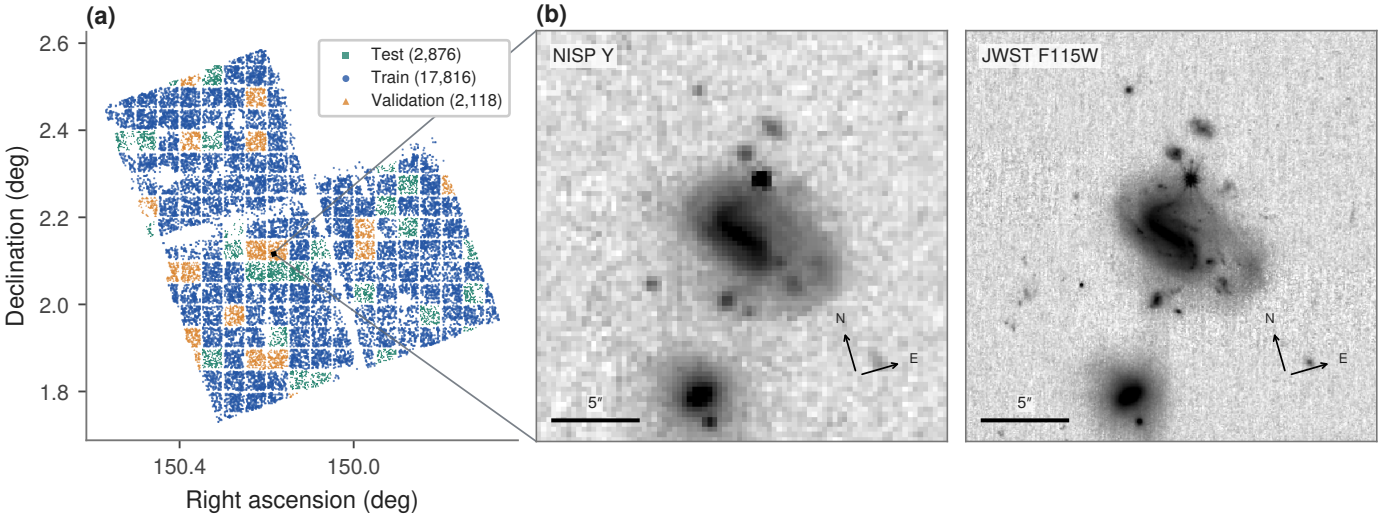
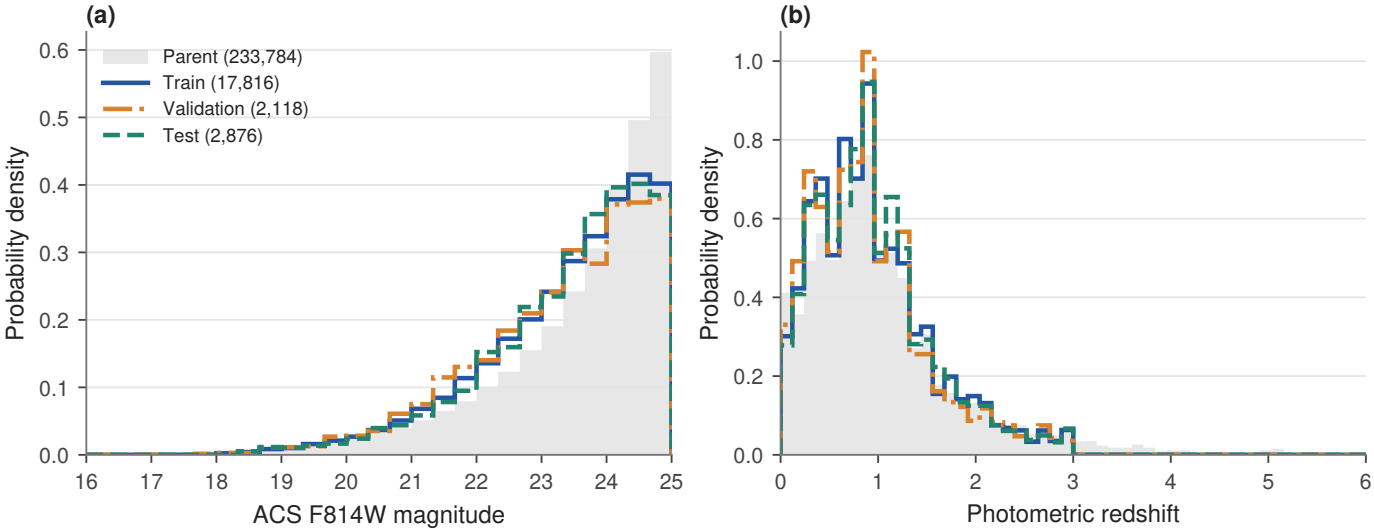
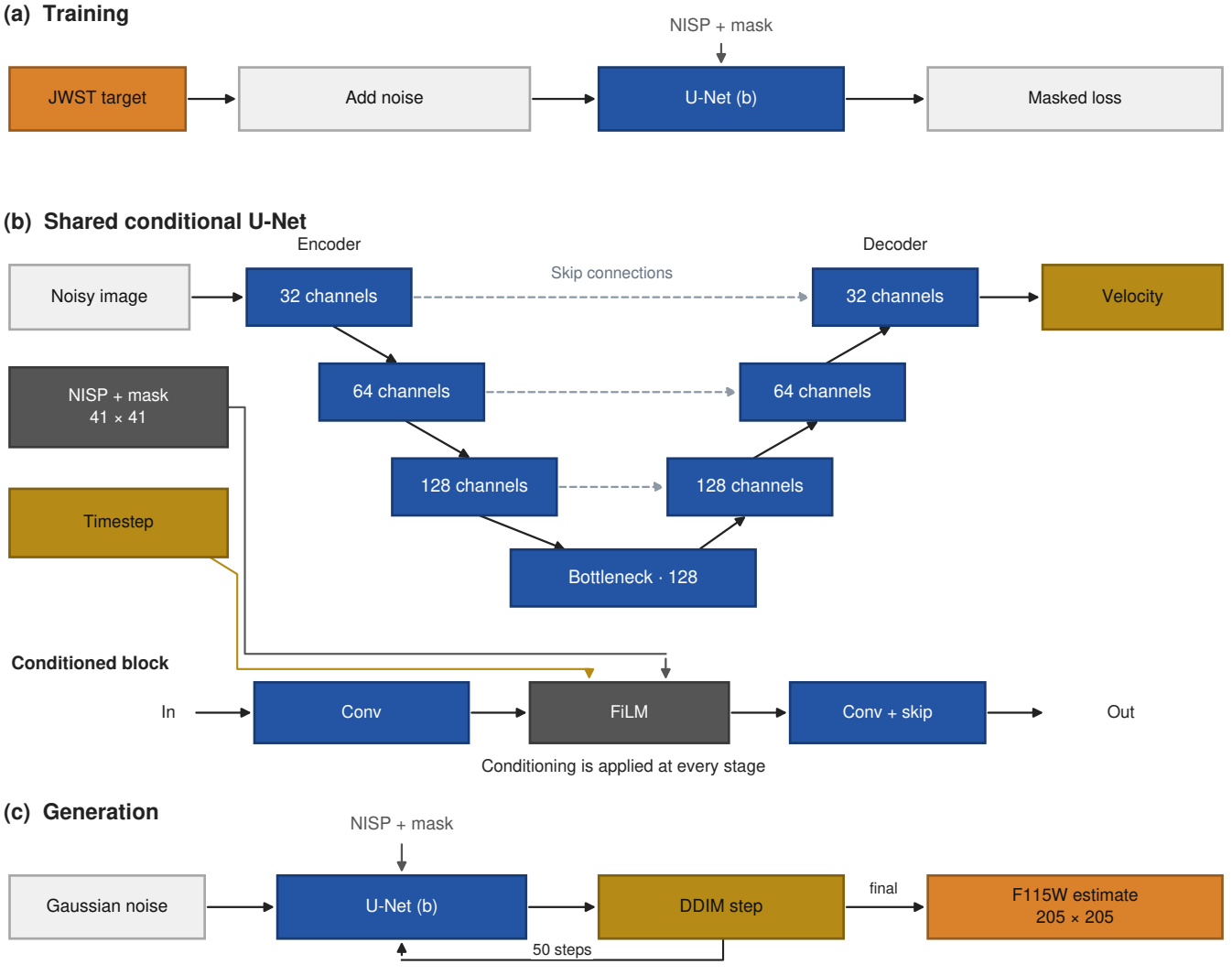
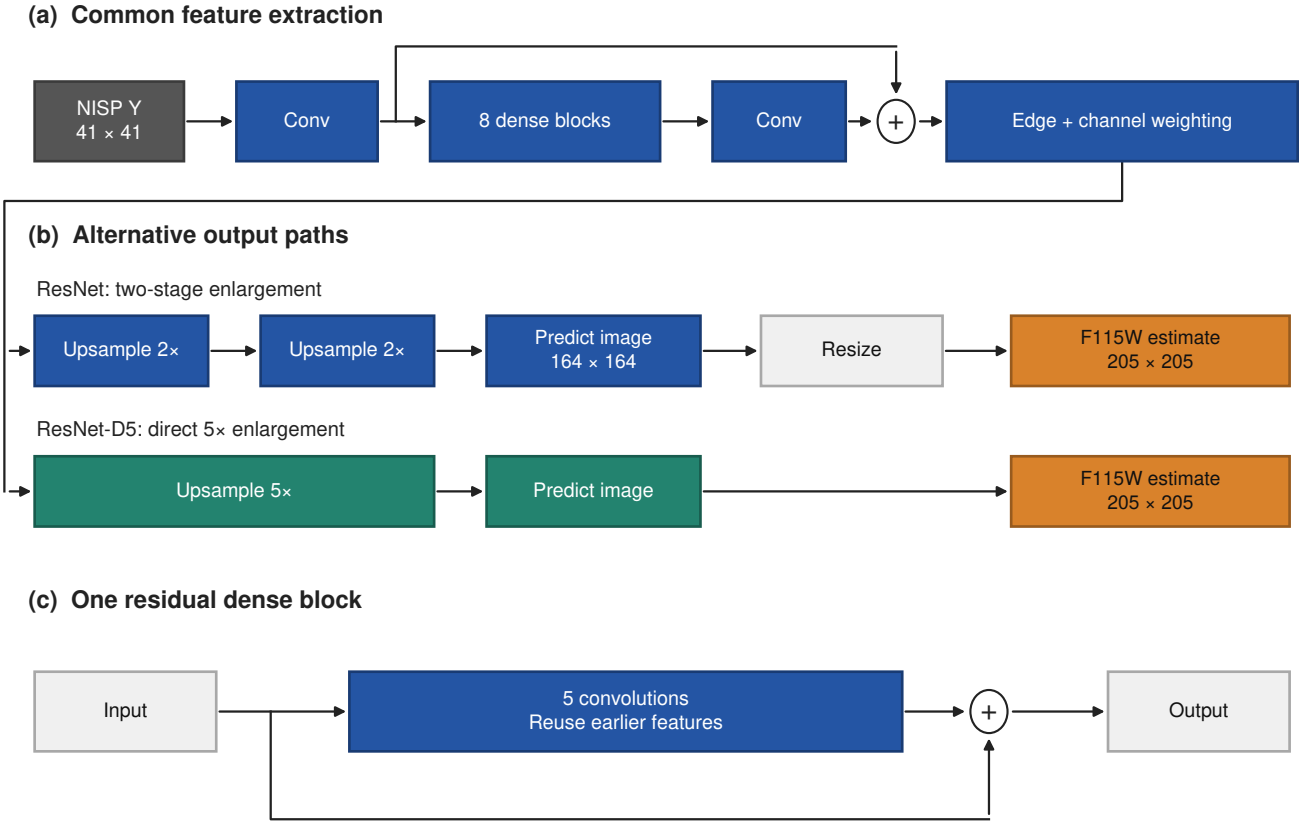
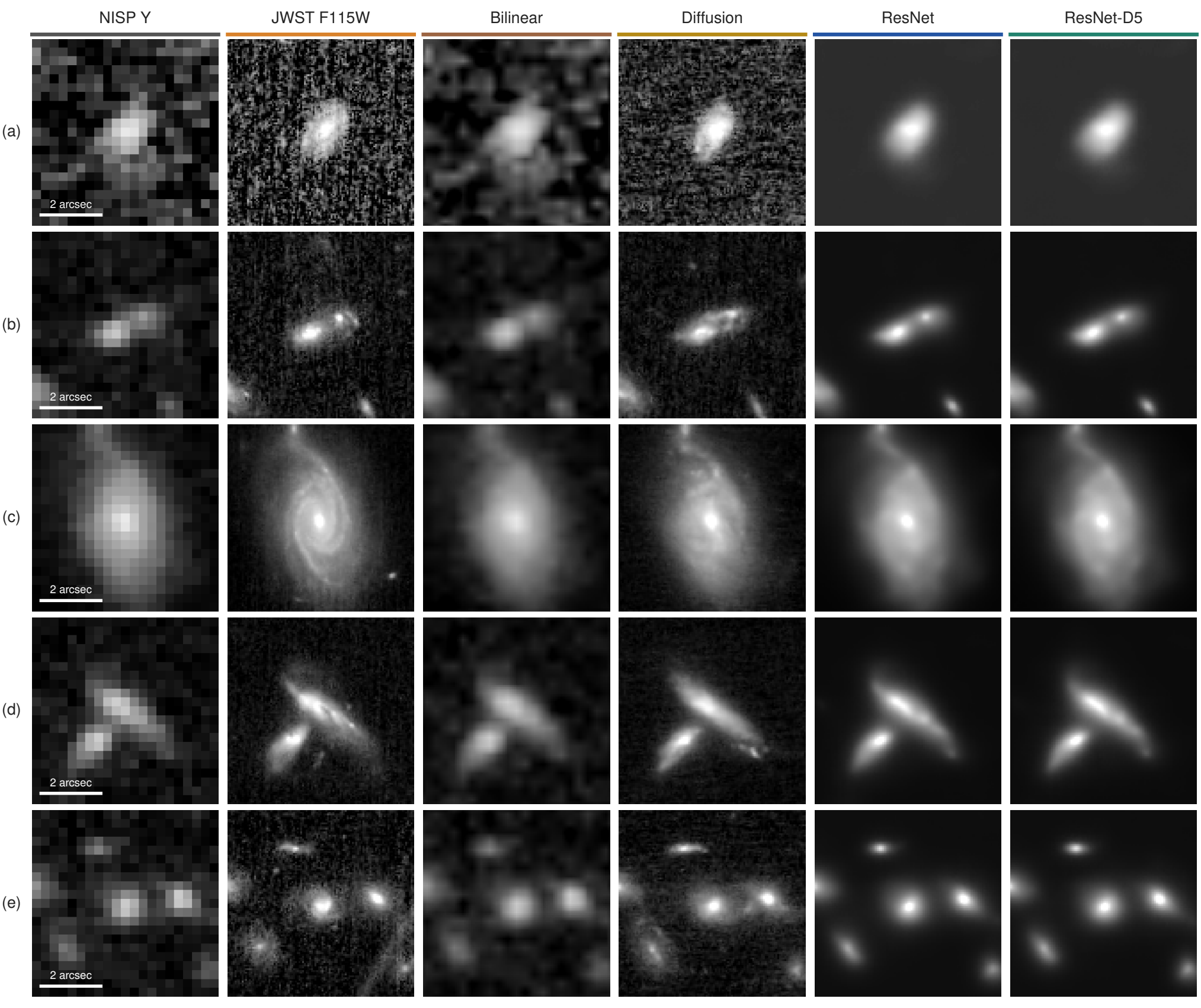
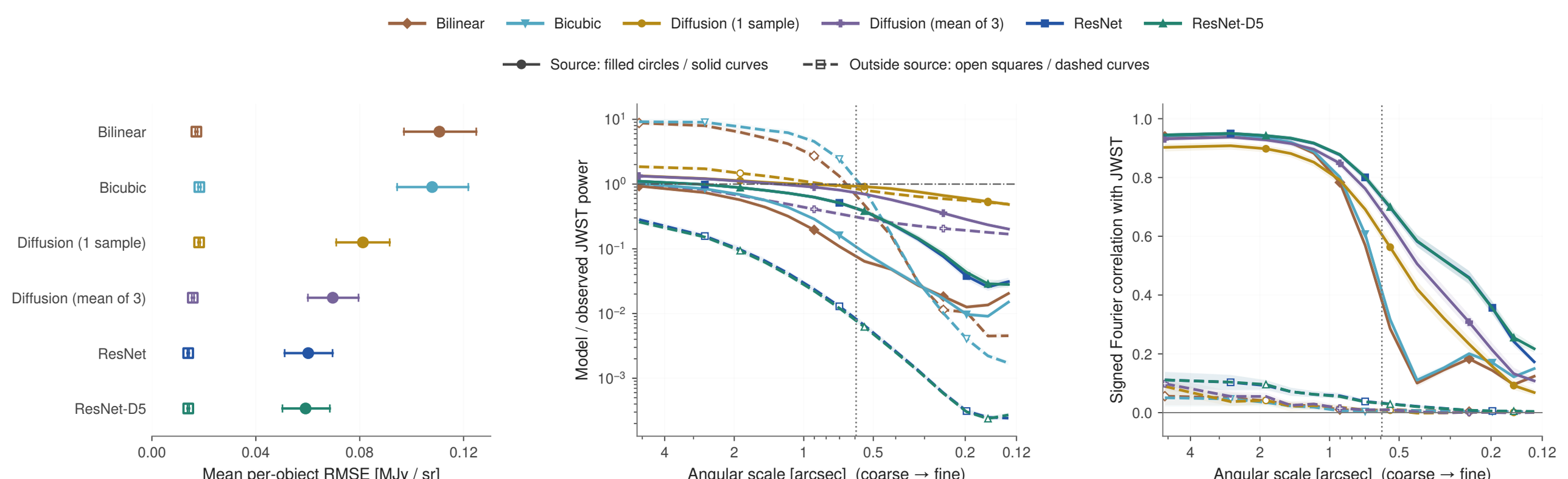

(a)
(b)
Test (2,876)
Train (17,816)
Validation (2,118)
Declination (deg)
2.6
2.4
2.2
2.0
1.8
150.4
150.0
Right ascension (deg)
NISP Y
JWST F115W
N
E
5″

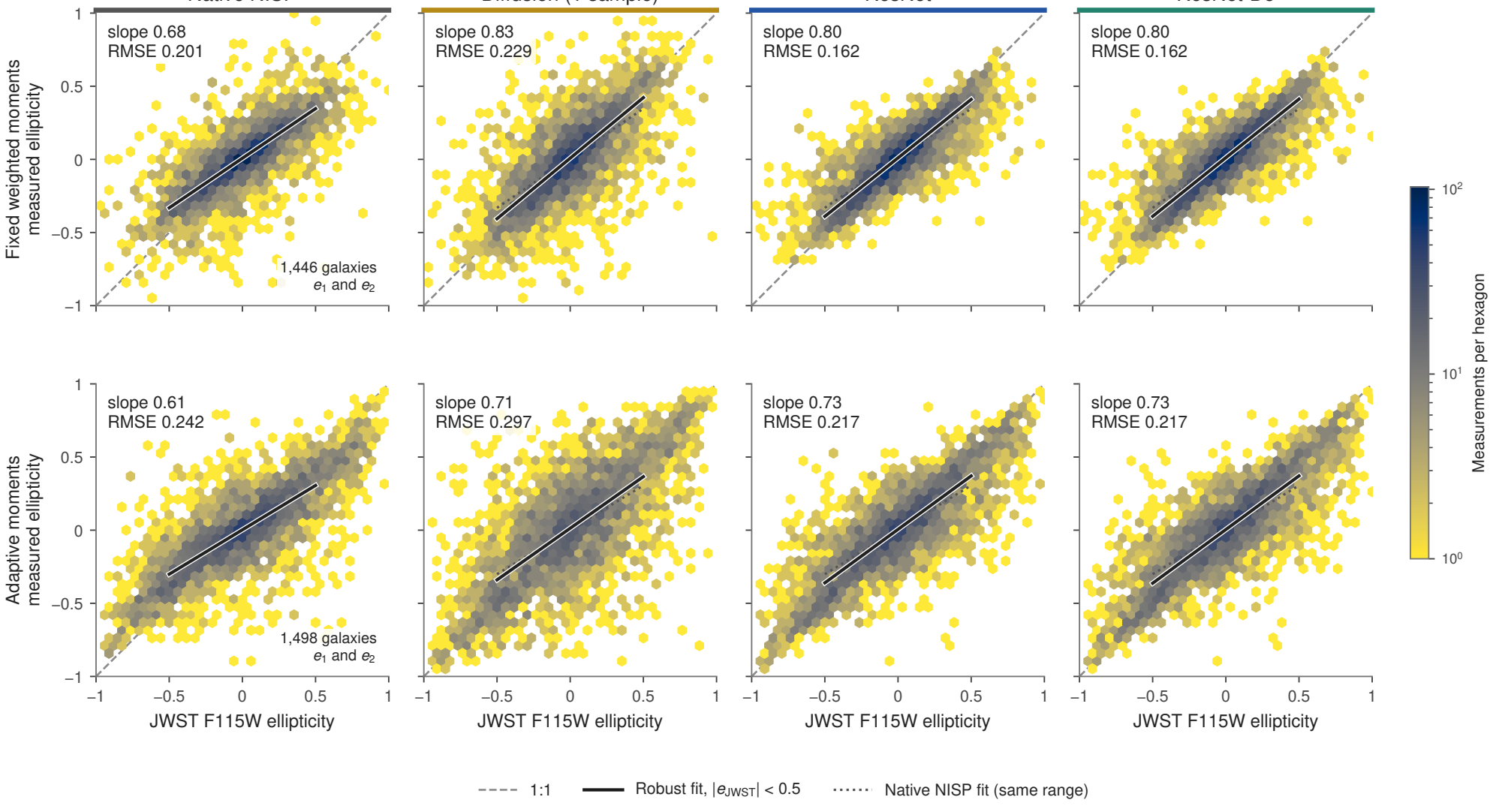

(a)
Parent (233,784)
Train (17,816)
Validation (2,118)
Test (2,876)
Probability density
0.0
0.1
0.2
0.3
0.4
0.5
0.6
16
17
18
19
20
21
22
23
24
25
ACS F814W magnitude
(b)
Probability density
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5
6
Photometric redshift

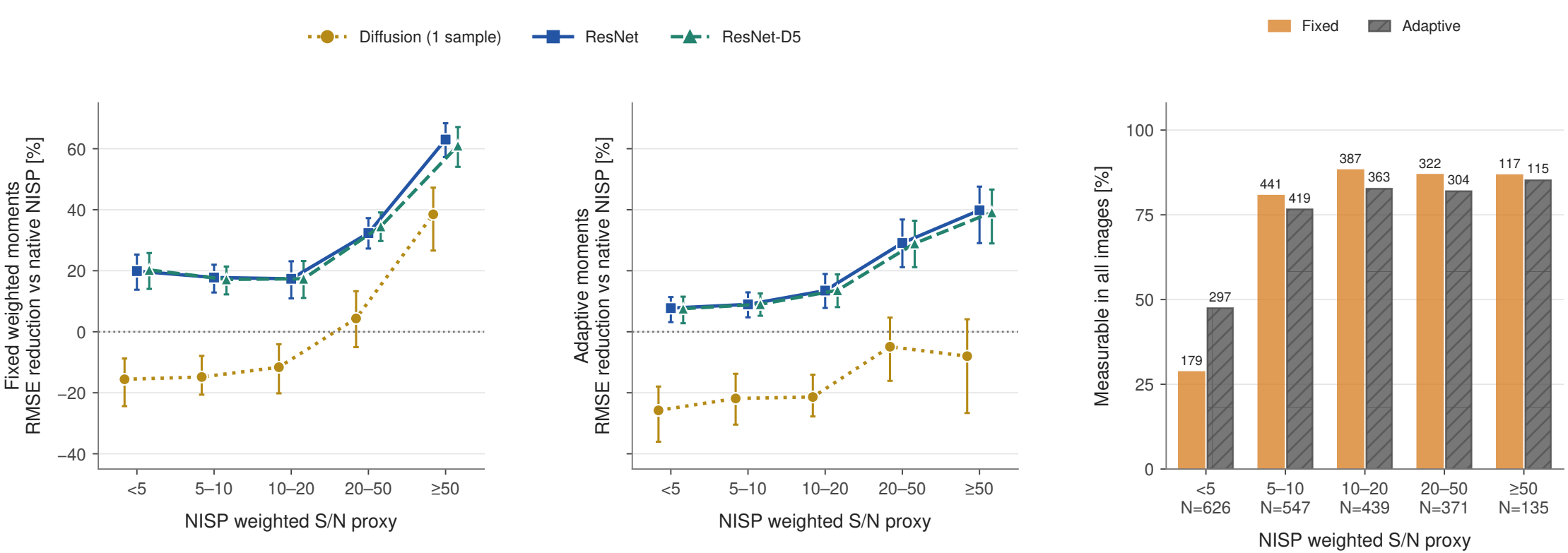

(a) Training
NISP + mask
JWST target
Add noise
U-Net (b)
Masked loss
(b) Shared conditional U-Net
Encoder
Decoder
Noisy image
32 channels
Skip connections
32 channels
Velocity
NISP + mask
41 × 41
64 channels
64 channels
128 channels
128 channels
Timestep
Bottleneck · 128
Conditioned block
In
Conv
FiLM
Conv + skip
Out
Conditioning is applied at every stage
(c) Generation
NISP + mask
Gaussian noise
U-Net (b)
DDIM step
final
F115W estimate
205 × 205
50 steps

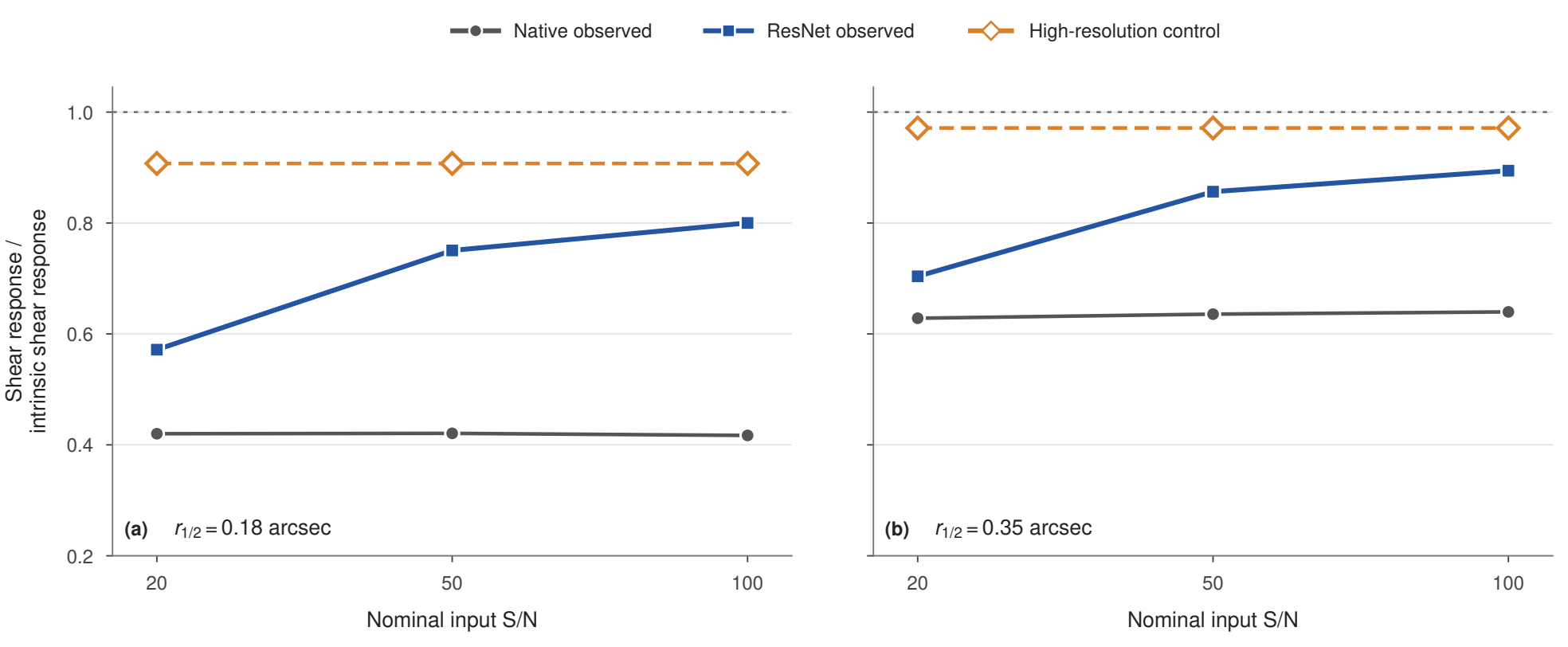

(a) Common feature extraction
NISP Y
41 × 41
Conv
8 dense blocks
Conv
+
Edge + channel weighting
(b) Alternative output paths
ResNet: two-stage enlargement
Upsample 2×
Upsample 2×
Predict image
164 × 164
Resize
F115W estimate
205 × 205
ResNet-D5: direct 5× enlargement
Upsample 5×
Predict image
F115W estimate
205 × 205
(c) One residual dense block
Input
5 convolutions
Reuse earlier features
+
Output


NISP Y
JWST F115W
Bilinear
Diffusion
ResNet
ResNet-D5
(a)
(b)
(c)
(d)
(e)
2 arcsec
2 arcsec
2 arcsec
2 arcsec
2 arcsec


Bilinear
Bicubic
Diffusion (1 sample)
Diffusion (mean of 3)
ResNet
ResNet-D5
Source: filled circles / solid curves
Outside source: open squares / dashed curves
Bilinear
Bicubic
Diffusion (1 sample)
Diffusion (mean of 3)
ResNet
ResNet-D5
0.00
0.04
0.08
0.12
Mean per-object RMSE [MJy / sr]
Model / observed JWST power
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
4
2
1
0.5
0.2
0.12
Angular scale [arcsec] (coarse → fine)
Signed Fourier correlation with JWST
1.0
0.8
0.6
0.4
0.2
0.0
4
2
1
0.5
0.2
0.12
Angular scale [arcsec] (coarse → fine)


Native NISP
Diffusion (1 sample)
ResNet
ResNet-D5
slope 0.68
RMSE 0.201
1,446 galaxies
e_1 and e_2
slope 0.83
RMSE 0.229
slope 0.80
RMSE 0.162
slope 0.80
RMSE 0.162
slope 0.61
RMSE 0.242
1,498 galaxies
e_1 and e_2
slope 0.71
RMSE 0.297
slope 0.73
RMSE 0.217
slope 0.73
RMSE 0.217
Fixed weighted moments measured ellipticity
Adaptive moments measured ellipticity
JWST F115W ellipticity
Measurements per hexagon
10^2
10^1
10^0
1:1
Robust fit, $|e_{JWST}| < 0.5$
Native NISP fit (same range)


Diffusion (1 sample)
ResNet
ResNet-D5
Fixed weighted moments
RMSE reduction vs native NISP [%]
60
40
20
0
−20
−40
<5
5–10
10–20
20–50
≥50
NISP weighted S/N proxy
Adaptive moments
RMSE reduction vs native NISP [%]
<5
5–10
10–20
20–50
≥50
NISP weighted S/N proxy
Fixed
Adaptive
Measurable in all images [%]
100
75
50
25
0
179
297
441
419
387
363
322
304
117
115
<5
N=626
5–10
N=547
10–20
N=439
20–50
N=371
≥50
N=135
NISP weighted S/N proxy


Native observed
ResNet observed
High-resolution control
Shear response / intrinsic shear response
1.0
0.8
0.6
0.4
0.2
(a) $r_{1/2} = 0.18$ arcsec
(b) $r_{1/2} = 0.35$ arcsec
20
50
100
Nominal input S/N
Nominal input S/N